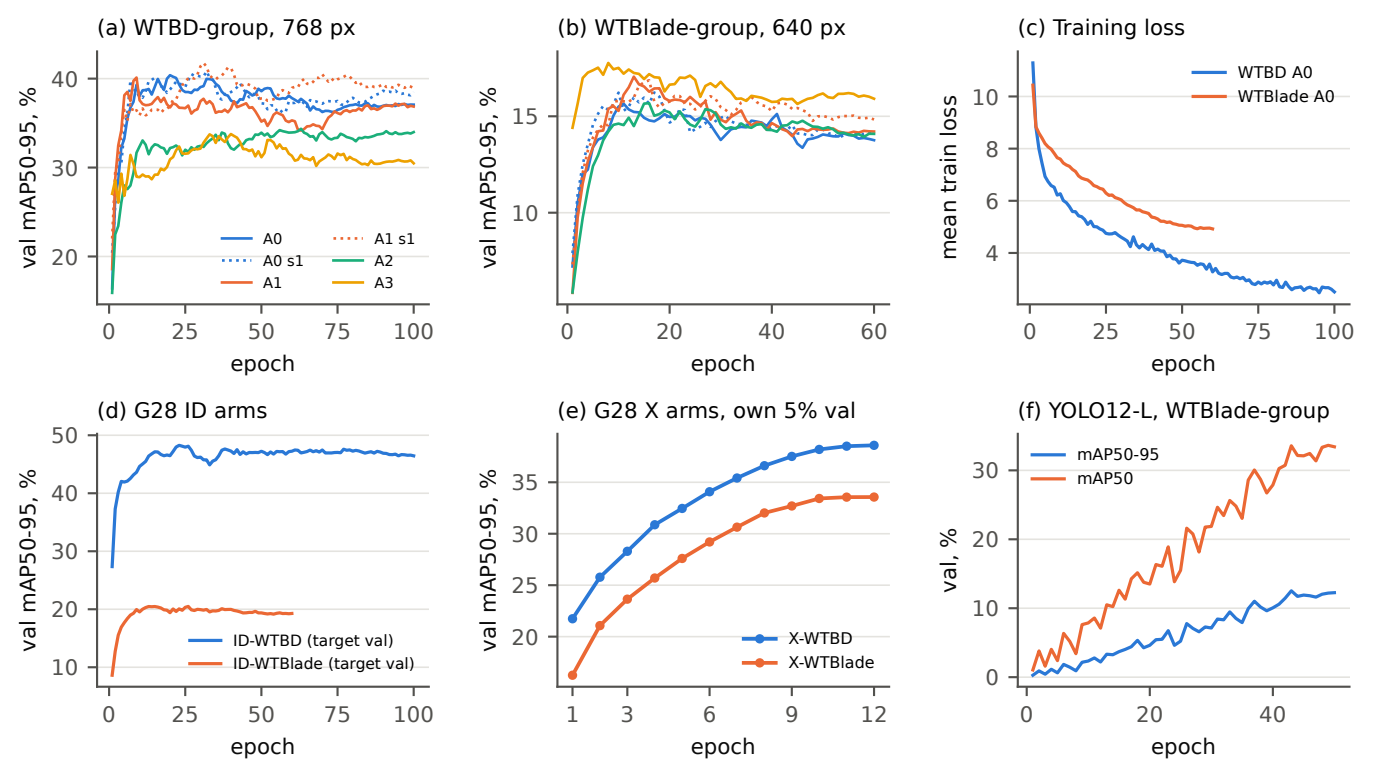
(a) WTBD-group, 768 px
val mAP50-95, %
epoch
A0
A0 s1
A1
A1 s1
A2
A3
(b) WTBlade-group, 640 px
val mAP50-95, %
epoch
(c) Training loss
mean train loss
epoch
WTBD A0
WTBlade A0
(d) G28 ID arms
val mAP50-95, %
epoch
ID-WTBD (target val)
ID-WTBlade (target val)
(e) G28 X arms, own 5% val
val mAP50-95, %
epoch
X-WTBD
X-WTBlade
(f) YOLO12-L, WTBlade-group
val, %
epoch
mAP50-95
mAP50